

July 29, 2026

RE: Staff, Applied AI Engineer

Dear Headstart Hiring Team,

I am excited to apply for the Staff, Applied AI Engineer role at Headstart. The company's model—learning how an enterprise actually works, reframing the problem with its leaders, and then shipping a production AI system that changes the operation—matches how I prefer to build. I am most effective where the technical path is initially ambiguous, the stakes are real, and success requires equal parts discovery, product judgment, hands-on engineering, and trust.

At RBC, I am building ASB Assist, an agentic GenAI layer that turns natural-language business requirements into compliant architecture diagrams and enterprise service recommendations. The work combines LLM + RAG orchestration, tool integration, prompt and context engineering, offline evaluation, grounding controls, auditability, and security patterns for a regulated environment. I treat AI as one tool in the system: the underlying ASB Builder platform remains a deterministic, rules-based product where reliability and explainability matter more than adding a model.

I also designed and led the 0-to-1 production delivery of that underlying full-stack platform using Node.js, TypeScript, React, and REST APIs, writing nearly all of its production implementation and core logic. I work directly with business leaders, product, risk, security, and principal architects to turn loosely framed needs into a tractable scope, explain trade-offs to executives, and decide which capabilities belong on the roadmap. Alongside that product work, I architected a distributed telemetry platform ingesting 150 TB per day across more than 10,000 systems, giving me a strong grounding in data pipelines, platform reliability, APIs, observability, and high-scale production operations.

The founding-engineer dimension of this role also resonates with me. As HealthCard's first full-stack engineer, I built the product from scratch across React/TypeScript, Node.js APIs, AWS EKS, verification services, applied ML, and HIPAA-aligned audit controls. The organization grew to more than 20 engineers before acquisition. Earlier freelance work required me to scope engagements directly with clients, translate business needs into architecture and delivery plans, handle difficult trade-offs, and own the outcome rather than only the implementation.

At staff level, I lead five engineers across concurrent AI, platform, governance, and observability initiatives. I stay hands-on through prototyping, architecture decisions, code and design reviews, production handoff, and incident-oriented thinking, while mentoring others and raising the quality bar across teams. My recent prototypes also use Python/FastAPI, React, PostgreSQL/pgvector, and LangGraph for agentic workflows with human approvals, evaluations, tracing, and audit logs. I would bring that same ownership to Headstart's client engagements and platform: move quickly during discovery, use AI when it creates real leverage, build the non-AI parts with equal care, and leave behind systems that clients can trust and maintain.

Thank you for your consideration. I would welcome the opportunity to discuss how my background in applied AI, full-stack product delivery, founding-engineer execution, distributed systems, and executive-facing technical leadership could contribute to Headstart.

Best regards,
Dhruv Doshi